

Publication List

Kevin Buck

July 23, 2026

Accepted Publications

- [1] Kevin Buck and Roger Temam. “Convergence Properties of PINNs for the Navier-Stokes-Cahn-Hilliard System”. In: *Pure and Applied Functional Analysis* (2025). To appear. arXiv: [2505.07964](https://arxiv.org/abs/2505.07964) [math.NA]. URL: <https://arxiv.org/abs/2505.07964>.
- [2] Kevin Buck and Woojeong Kim. “Auto-Adaptive PINNs with Applications to Phase Transitions”. In: *Numerical Mathematics: Theory, Methods and Applications* (2026). To appear. arXiv: [2510.23999](https://arxiv.org/abs/2510.23999) [math.NA]. URL: <https://arxiv.org/abs/2510.23999>.

Submitted Publications

- [3] Kevin Buck et al. *Nonconvex optimization and convergence of stochastic gradient descent, and solution of asynchronous game*. submitted. 2025. arXiv: [2503.02155](https://arxiv.org/abs/2503.02155) [math.OC]. URL: <https://arxiv.org/abs/2503.02155>.

Other Preprints

- [4] Kevin Buck et al. *Continuous guts poker and numerical optimization of generalized recursive games*. 2022. arXiv: [2208.02788](https://arxiv.org/abs/2208.02788) [math.OC]. URL: <https://arxiv.org/abs/2208.02788>.
- [5] Jessica Babyak et al. *Synchronous vs. asynchronous coalitions in multiplayer games, with applications to guts poker*. 2024. arXiv: [2412.19855](https://arxiv.org/abs/2412.19855) [cs.GT]. URL: <https://arxiv.org/abs/2412.19855>.